HCR240
PATHOPHYSIOLOGY
-ARIZONA COLLEGE-

MODULE 2 & MODULE 3

NOTES

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1. Disorders of the Immune System

Immune disorders develop when immune regulation becomes abnormal.

Major immune disease categories include:

- autoimmunity
- hypersensitivity
- immunodeficiency ^^

The immune system must distinguish:

- self
  from
- non-self

Failure of immune regulation
→ autoreactive lymphocytes survive
→ inappropriate immune activation develops
→ inflammation and tissue injury occur

1.1 Immune Tolerance

Immune tolerance prevents the immune system from attacking normal self-tissue.

Developing lymphocytes generate random antigen receptors.

Random receptor generation is necessary because the immune system must recognize many different pathogens.

However:

Random receptor generation
→ some lymphocytes recognize self-antigens
→ potentially dangerous autoreactive lymphocytes form

Tolerance mechanisms therefore eliminate or suppress self-reactive lymphocytes.

Tolerance occurs through:

- central tolerance
- peripheral tolerance

1.2 Central Tolerance

Central tolerance occurs during lymphocyte development.

For T cells, this occurs primarily within the thymus.

Developing T cells undergo:

- positive selection
- negative selection

1.2A Positive Selection

Positive selection tests whether immature T cells can recognize MHC molecules.

Mechanism:

Immature T cell enters thymus
→ thymic epithelial cells display self-MHC
→ T-cell receptor attempts MHC binding

No MHC recognition
→ apoptosis occurs

Reason:

T cells must recognize antigen presented on MHC molecules.

No MHC recognition
→ T cell cannot function
→ nonfunctional T cell removed

Weak or moderate MHC recognition
→ T cell survives
→ positive selection successful

Positive selection therefore preserves functional T cells.

1.2B Negative Selection

Negative selection tests whether T cells react strongly against self-antigens.

Mechanism:

Surviving T cells encounter self-antigens in thymus
→ strong self-antigen binding occurs
→ apoptosis triggered
→ autoreactive T cell destroyed

Strong self-recognition is dangerous because the T cell may later attack normal tissue.

Negative selection therefore removes dangerous autoreactive lymphocytes.

Failure of negative selection
→ autoreactive T cells survive
→ autoimmune disease risk increases

1.3 Peripheral Tolerance

Some autoreactive lymphocytes escape central tolerance.

Peripheral tolerance suppresses these cells outside the thymus and bone marrow.

Mechanisms include:

- anergy
- regulatory T cells
- apoptosis

1.3A Anergy

Anergy means functional inactivation.

T-cell activation normally requires:

- antigen recognition
- co-stimulation

If antigen recognition occurs without proper co-stimulation:

T cell recognizes antigen
→ incomplete activation occurs
→ T cell becomes inactive
→ tissue attack prevented

1.3B Regulatory T Cells

Regulatory T cells suppress excessive immune activation.

Mechanism:

Regulatory T cell recognizes excessive immune response
→ suppressive cytokines released
→ autoreactive lymphocyte activity decreases
→ tissue injury limited

1.4 Autoimmunity ^^

Autoimmunity develops when tolerance mechanisms fail. ^^

Mechanism:

Failure of central or peripheral tolerance
→ autoreactive lymphocytes survive
→ self-antigens recognized as dangerous
→ cytokine release and antibody production occur
→ chronic inflammation develops
→ tissue injury occurs

Autoimmune disease therefore represents loss of immune regulation rather than “strong immunity.”

1.5 Systemic Lupus Erythematosus (SLE) ^^

SLE is a systemic autoimmune disease associated with immune-complex deposition. ^^

Mechanism:

Failure of immune tolerance
→ autoreactive B cells survive
→ autoantibodies form against nuclear antigens
→ antigen-antibody immune complexes form
→ immune complexes circulate
→ immune complexes deposit in tissues
→ complement activated
→ neutrophils recruited
→ inflammatory tissue injury develops

Important autoantibodies include:

- ANA ^^
- anti-dsDNA ^^

1.5A SLE and Joint Pain ^^

Immune complexes deposit in synovial tissue.

Immune-complex deposition
→ complement activation
→ neutrophil recruitment
→ inflammatory cytokine release
→ synovial inflammation develops
→ joint stiffness and pain occur ^^

1.5B SLE and Butterfly Rash ^^

Immune complexes deposit within dermal blood vessels.

Immune-complex deposition in skin
→ complement activation
→ vascular inflammation develops
→ tissue injury occurs
→ malar butterfly rash develops across cheeks and nose ^^

Photosensitivity worsens inflammation because UV injury increases cellular damage and antigen exposure.

1.5C SLE and Hypertension ^^

Immune complexes deposit in glomeruli.

Immune-complex deposition in glomeruli
→ glomerulonephritis develops
→ glomerular filtration decreases
→ sodium and water retention increase
→ intravascular volume increases
→ hypertension develops ^^

Proteinuria develops because inflammatory glomerular injury increases protein leakage through the filtration barrier.

1.5D SLE Clinical Findings ^^

Common manifestations include:

- fatigue
- fever ^^
- painful or stiff joints ^^
- butterfly rash ^^
- photosensitivity
- renal dysfunction
- hypertension from renal involvement ^^
- positive ANA ^^
- positive anti-dsDNA ^^

SLE is classified as Type III hypersensitivity because tissue injury is caused by immune-complex deposition. ^^

1.6 Rheumatoid Arthritis (RA) ^^

Rheumatoid arthritis is chronic autoimmune inflammatory joint disease. ^^

Mechanism:

Failure of immune tolerance
→ autoreactive lymphocytes target synovial tissue
→ cytokine release develops
→ chronic synovial inflammation occurs
→ pannus forms
→ cartilage destruction develops
→ bone erosion develops
→ joint deformity occurs

Important cytokines include:

- TNF-alpha
- IL-1
- IL-6

Pannus is inflammatory granulation tissue growing across the joint surface.

Pannus formation
→ cartilage erosion
→ narrowing of joint space
→ pain and stiffness worsen
→ deformity develops

RA typically affects joints symmetrically.

Common findings include:

- morning stiffness
- swollen joints
- painful joints
- decreased range of motion
- chronic hand deformities

1.7 Hypersensitivity Reactions ^^

Hypersensitivity reactions are exaggerated or misdirected immune responses causing tissue injury. ^^

The four major hypersensitivity reactions are:

- Type I hypersensitivity ^^
- Type II hypersensitivity ^^
- Type III hypersensitivity ^^
- Type IV hypersensitivity ^^

1.8 Type I Hypersensitivity ^^

Type I hypersensitivity is immediate hypersensitivity mediated by IgE and mast cells. ^^

Mechanism:

Initial allergen exposure
→ helper T-cell activation occurs
→ B-cell class switching to IgE occurs
→ IgE binds mast-cell Fc receptors
→ sensitization develops

Re-exposure to same allergen
→ allergen cross-links mast-cell-bound IgE
→ mast-cell degranulation occurs
→ histamine released
→ immediate allergic symptoms develop

Histamine causes:

- vasodilation
- increased vascular permeability
- itching
- mucus production
- bronchoconstriction

Important mediators and cells include:

- IgE ^^
- mast cells ^^

1.8A Allergic Rhinitis and Hay Fever ^^

Allergen exposure
→ mast-cell activation
→ histamine release
→ vascular permeability increases
→ mucus secretion increases
→ nasal inflammation develops ^^

Common findings include:

- rhinorrhea
- sneezing
- itching
- congestion

Hay fever ^^
and
allergic rhinitis ^^
are Type I hypersensitivity reactions.

1.8B Urticaria ^^

Histamine release within superficial dermis
→ vascular permeability increases
→ localized edema develops
→ raised pruritic wheals form

Urticaria ^^
is commonly called hives.

1.8C Anaphylaxis ^^

Anaphylaxis is severe systemic Type I hypersensitivity. ^^

Mechanism:

Systemic allergen exposure
→ widespread mast-cell degranulation
→ massive histamine release
→ systemic vasodilation develops
→ vascular permeability increases
→ hypotension develops
→ bronchoconstriction develops
→ airway edema develops
→ respiratory compromise may occur

Epinephrine treatment:

Epinephrine
→ vasoconstriction increases blood pressure
→ bronchodilation improves airflow

1.9 Type II Hypersensitivity ^^

Type II hypersensitivity occurs when antibodies bind directly to cell-surface antigens. ^^

Primary antibodies involved:

- IgG
- IgM

Mechanism:

Antibody binds cell-surface antigen
→ complement activated
→ inflammatory destruction occurs
→ target cells damaged

1.9A Transfusion Reaction ^^

Incompatible blood transfusion
→ recipient antibodies bind donor RBC antigens
→ complement activated
→ RBC hemolysis develops
→ inflammatory reaction occurs ^^

Hemolysis releases hemoglobin into circulation.

Possible findings include:

- fever
- chills
- hypotension
- hemoglobinuria
- renal injury

1.9B Hemolytic Disease of the Newborn ^^

Rh-negative mother exposed to Rh-positive fetal blood
→ maternal anti-Rh antibodies form
→ maternal IgG crosses placenta during later pregnancy
→ fetal RBC destruction develops
→ fetal hemolytic anemia occurs ^^

RBC destruction
→ bilirubin production increases
→ jaundice develops

Severe bilirubin accumulation may cause kernicterus.

Phototherapy ^^
helps reduce bilirubin accumulation.

1.10 Type III Hypersensitivity ^^

Type III hypersensitivity occurs when immune complexes deposit within tissues. ^^

Mechanism:

Antigen-antibody complexes form
→ immune complexes circulate
→ immune complexes deposit in tissues
→ complement activated
→ neutrophils recruited
→ inflammatory tissue injury develops

Common deposition sites include:

- kidneys
- joints
- skin
- blood vessels

SLE ^^
is a classic Type III hypersensitivity disease.

1.11 Type IV Hypersensitivity ^^

Type IV hypersensitivity is delayed T-cell-mediated hypersensitivity. ^^

Antibodies are not the primary mechanism.

Mechanism:

Antigen exposure
→ sensitized T cells form
→ re-exposure activates memory T cells
→ cytokine release occurs
→ macrophages activated
→ delayed inflammatory injury develops

Type IV reactions usually develop over 24–72 hours.

Examples include:

- poison ivy ^^
- contact dermatitis ^^
- tuberculosis skin test ^^

1.12 Immunodeficiency ^^

Immunodeficiency occurs when immune defense mechanisms are impaired. ^^

Mechanism:

Defective immune cells
or
Defective antibodies
or
Defective immune signaling
→ poor pathogen clearance develops
→ recurrent infections occur

Immunodeficiency disorders may affect:

- T cells
- B cells
- antibodies

- combined immunity

1.13 Severe Combined Immunodeficiency Disease (SCID) ^^

SCID is severe combined impairment of both cellular and humoral immunity. ^^

Mechanism:

Genetic defect impairs lymphocyte development
→ T-cell maturation impaired
→ helper T-cell function decreases
→ B-cell activation impaired
→ antibody responses weaken
→ severe adaptive immune dysfunction develops

Impaired T-cell function
→ viral and fungal defense weakened

Impaired B-cell activation
→ bacterial defense weakened

Common findings include:

- recurrent respiratory infections ^^
- recurrent pneumonia ^^
- recurrent bronchitis ^^
- opportunistic infections
- failure to thrive
- frequent antibiotic treatment ^^

1.14 Selective IgA Deficiency ^^

Selective IgA deficiency is decreased IgA production. ^^

IgA normally protects mucosal surfaces.

Reduced IgA production
→ decreased mucosal immunity

→ pathogens attach more easily to respiratory and GI mucosa
→ recurrent mucosal infections develop

Common findings include:

- recurrent sinus infections
- recurrent respiratory infections ^^
- recurrent GI infections
- chronic mucosal infections

Selective IgA deficiency ^^
is one of the most common primary immunodeficiency disorders.

1.15 Hypogammaglobulinemia ^^

Hypogammaglobulinemia involves decreased immunoglobulin production. ^^

Mechanism:

Reduced antibody production
→ impaired opsonization and neutralization
→ recurrent bacterial infections develop

Common findings include:

- recurrent respiratory infections
- chronic sinus infections
- recurrent pneumonia

1.16 Chronic Mucocutaneous Candidiasis ^^

Chronic mucocutaneous candidiasis involves persistent Candida infections affecting skin and mucosal surfaces. ^^

Mechanism:

T-cell dysfunction develops
→ impaired fungal immunity occurs
→ chronic Candida overgrowth develops

Common findings include:

- oral thrush
- chronic fungal skin infections
- mucosal Candida infections

1.17 DiGeorge Syndrome ^^

DiGeorge syndrome is associated with abnormal thymic development. ^^

Mechanism:

22q11 deletion
→ thymic hypoplasia develops
→ T-cell maturation impaired
→ cellular immunity weakened

Reduced T-cell function
→ impaired viral defense
→ impaired fungal defense

Associated findings may include:

- recurrent infections
- congenital heart defects
- hypocalcemia

1.18 HIV and AIDS ^^

HIV infects CD4 helper T cells. ^^

Mechanism:

HIV enters bloodstream
→ virus binds CD4 receptors
→ viral entry into helper T cells occurs
→ reverse transcriptase converts viral RNA into DNA
→ viral DNA integrates into host genome
→ viral replication develops
→ CD4 helper T-cell destruction occurs

Loss of helper T cells
→ impaired immune coordination develops
→ macrophage activation weakens
→ cytotoxic T-cell activation weakens
→ B-cell activation weakens
→ opportunistic infections increase

AIDS ^^
develops when severe CD4 depletion causes profound immunodeficiency.

2. Disorders of White Blood Cells and Hematologic Malignancy

White blood cells normally defend the body against infection, abnormal cells, and foreign material.

Major leukocyte groups include:

- neutrophils
- lymphocytes
- monocytes/macrophages
- eosinophils
- basophils

Normal hematopoiesis occurs in bone marrow.

Hematopoietic stem cell
→ myeloid stem cells and lymphoid stem cells form
→ mature blood cells produced
→ functional immune defense maintained

Disorders develop when:

- leukocyte production becomes malignant
- leukocyte maturation fails
- leukocyte numbers become abnormal
- leukocyte function becomes impaired

2.1 Leukocytosis ^^

Leukocytosis means elevated white blood cell count. ^^

Mechanism:

Infection or inflammation
→ cytokines stimulate bone marrow
→ leukocyte production increases
→ circulating WBC count rises

Different leukocyte elevations suggest different disease processes.

2.1A Neutrophilia ^^

Neutrophils are major phagocytic cells involved in acute bacterial inflammation.

Bacterial infection
→ macrophage cytokine release develops
→ bone marrow stimulation occurs
→ neutrophil production increases
→ neutrophilia develops ^^

Neutrophilia commonly suggests:

- bacterial infection ^^
- acute inflammation
- tissue necrosis
- stress response

2.1B Lymphocytosis ^^

Lymphocytes are central to adaptive immunity.

Viral infection
→ lymphocyte activation develops
→ lymphocyte proliferation increases
→ lymphocytosis develops ^^

Lymphocytosis commonly suggests:

- viral infection ^^
- infectious mononucleosis ^^
- chronic immune stimulation

2.1C Eosinophilia ^^

Eosinophils participate in allergic disease and parasitic defense.

Parasitic infection or allergic stimulation
→ IL-5 production increases

→ eosinophil production increases
→ eosinophilia develops ^^

Eosinophilia commonly suggests:

- parasitic infection ^^
- allergic disease ^^
- asthma
- hypersensitivity reactions

2.2 Infectious Mononucleosis ^^

Infectious mononucleosis is commonly caused by Epstein-Barr virus. ^^

Mechanism:

EBV infects B lymphocytes
→ infected B cells proliferate
→ cytotoxic T-cell response develops
→ lymphoid tissue enlarges
→ systemic inflammatory symptoms develop

Common findings include:

- fever
- sore throat
- lymphadenopathy ^^
- fatigue ^^
- atypical lymphocytes ^^

Atypical lymphocytes are reactive cytotoxic T cells responding to infected B cells.

2.3 Leukopenia ^^

Leukopenia means decreased white blood cell count. ^^

Mechanism:

Bone marrow suppression
or

Immune destruction
or
Viral suppression
→ leukocyte numbers decrease
→ infection risk increases

Causes may include:

- chemotherapy ^^
- radiation exposure
- severe viral infection
- bone marrow failure

Reduced leukocyte count
→ impaired pathogen defense
→ opportunistic infection risk increases

2.4 Leukemia ^^

Leukemia is malignant proliferation of white blood cell precursors within bone marrow. ^^

Mechanism:

DNA mutation develops in hematopoietic precursor cell
→ malignant clone forms
→ uncontrolled leukocyte proliferation develops
→ abnormal leukocytes accumulate in marrow and blood

Bone marrow overcrowding develops.

Malignant leukocyte expansion
→ normal hematopoiesis suppressed
→ anemia develops
→ thrombocytopenia develops
→ infection risk increases

Leukemia therefore often produces:

- fatigue from anemia
- bleeding from thrombocytopenia
- infection from dysfunctional leukocytes

2.4A Acute Leukemia ^^

Acute leukemia involves immature blast cells. ^^

Mechanism:

Mutation in immature precursor cell
→ blast-cell proliferation develops
→ maturation failure occurs
→ immature nonfunctional leukocytes accumulate rapidly

Because blast cells fail to mature:

Large numbers of dysfunctional cells accumulate
→ normal marrow function collapses rapidly
→ severe symptoms develop quickly

Common findings include:

- fatigue ^^
- bleeding ^^
- recurrent infections ^^
- bone pain
- blast cells in blood or marrow ^^

Acute leukemia progresses rapidly.

2.4B Chronic Leukemia ^^

Chronic leukemia involves more mature malignant leukocytes. ^^

Mechanism:

Malignant transformation occurs in more differentiated leukocyte lineage
→ malignant cells continue partial maturation
→ slower accumulation develops

Because cells are somewhat more mature:

Disease progression is often slower initially.

Patients may initially have:

- leukocytosis ^^
- fatigue
- splenomegaly
- weight loss

2.5 Acute Lymphoblastic Leukemia (ALL) ^^

ALL is malignant proliferation of lymphoblasts. ^^

Mechanism:

Mutation develops in lymphoid precursor cell
→ lymphoblast proliferation develops
→ marrow overcrowding occurs
→ normal blood-cell production suppressed

ALL commonly occurs in children. ^^

Bone marrow failure develops because malignant blasts replace normal marrow tissue.

Consequences include:

- anemia
- thrombocytopenia
- infection risk
- fatigue ^^
- bleeding ^^

2.6 Acute Myelogenous Leukemia (AML) ^^

AML is malignant proliferation of myeloid precursor cells. ^^

Mechanism:

Mutation develops in myeloid precursor
→ myeloblast proliferation develops
→ marrow failure develops
→ dysfunctional leukocytes accumulate

Auer rods may be seen in myeloblasts. ^^

AML commonly occurs in adults. ^^

2.7 Chronic Lymphocytic Leukemia (CLL) ^^

CLL involves malignant proliferation of mature-appearing B lymphocytes. ^^

Mechanism:

Malignant B-cell clone develops
→ slow lymphocyte accumulation occurs
→ abnormal lymphocytes circulate in blood and marrow

Common findings include:

- lymphocytosis ^^
- lymphadenopathy ^^
- splenomegaly
- fatigue

Smudge cells may appear on peripheral smear. ^^

2.8 Chronic Myelogenous Leukemia (CML) ^^

CML is associated with BCR-ABL fusion gene formation. ^^

Mechanism:

Chromosomal translocation develops
→ Philadelphia chromosome forms ^^
→ BCR-ABL tyrosine kinase produced
→ uncontrolled myeloid proliferation develops

BCR-ABL continuously stimulates growth signaling.

Consequences:

Excessive granulocyte production
→ marked leukocytosis develops
→ splenomegaly develops
→ hypermetabolic symptoms develop

Common findings include:

- fatigue
- weight loss
- night sweats
- leukocytosis ^^
- splenomegaly

Tyrosine kinase inhibitors target abnormal BCR-ABL signaling.

2.9 Lymphoma ^^

Lymphoma is malignant proliferation of lymphoid tissue. ^^

Unlike leukemia, lymphoma primarily forms solid masses in lymphoid tissue.

Mechanism:

Mutation develops in lymphocyte
→ malignant lymphocyte clone proliferates
→ lymph-node architecture replaced
→ lymphatic masses enlarge

Common findings include:

- lymphadenopathy ^^
- fever
- night sweats ^^
- weight loss ^^

These are called “B symptoms.”

2.9A Hodgkin Lymphoma ^^

Hodgkin lymphoma is associated with Reed-Sternberg cells. ^^

Mechanism:

Malignant B-cell transformation develops
→ abnormal giant multinucleated cells form

→ inflammatory background develops
→ progressive lymph-node enlargement occurs

Reed-Sternberg cells ^^
are classically associated with Hodgkin lymphoma.

Lymph-node spread is often orderly and contiguous.

2.9B Non-Hodgkin Lymphoma ^^

Non-Hodgkin lymphoma includes multiple malignant lymphoid neoplasms without Reed-Sternberg cells. ^^

Spread is often less predictable than Hodgkin lymphoma.

Mechanism:

Malignant lymphocyte clone develops
→ lymphoid tissue proliferation occurs
→ extranodal spread may occur

2.10 Multiple Myeloma ^^

Multiple myeloma is malignant proliferation of plasma cells. ^^

Plasma cells normally produce antibodies.

Mechanism:

Malignant plasma-cell clone develops
→ monoclonal immunoglobulin production increases
→ plasma cells accumulate in marrow
→ bone destruction develops

Malignant plasma cells stimulate osteoclast activity.

Osteoclast activation
→ bone resorption increases
→ lytic bone lesions develop
→ fractures and bone pain occur ^^

Bone destruction releases calcium.

Bone resorption
→ hypercalcemia develops ^^

Abnormal immunoglobulins may damage kidneys.

Monoclonal protein accumulation
→ renal injury develops ^^

Common findings include:

- bone pain ^^
- lytic lesions ^^
- hypercalcemia ^^
- renal dysfunction ^^
- recurrent infections
- monoclonal spike on electrophoresis ^^

3. Disorders of Red Blood Cells and Oxygen Transport

Red blood cells transport oxygen to tissues.

Normal oxygen delivery requires:

- adequate RBC production
- adequate hemoglobin
- normal iron availability
- normal bone marrow function
- normal RBC survival
- adequate circulation

Hemoglobin inside RBCs binds oxygen in the lungs and delivers oxygen to tissues.

Reduced RBC function
or
Reduced hemoglobin
→ tissue oxygen delivery decreases
→ cellular hypoxia develops
→ fatigue and organ dysfunction occur

3.1 Erythropoiesis

Red blood cells are produced in bone marrow.

Mechanism:

Renal oxygen sensing detects hypoxia
→ erythropoietin release increases
→ bone marrow stimulated
→ erythrocyte production increases

Erythropoietin is primarily produced by the kidneys.

Chronic kidney disease
→ decreased erythropoietin production

→ decreased RBC production
→ anemia develops ^^

3.2 Anemia ^^

Anemia means reduced oxygen-carrying capacity of blood. ^^

Anemia may result from:

- blood loss
- decreased RBC production
- increased RBC destruction

Reduced oxygen delivery
→ tissue hypoxia develops
→ compensatory physiologic responses occur

Compensatory responses include:

- tachycardia
- increased respiratory rate
- fatigue
- weakness

Cells shift toward anaerobic metabolism when oxygen delivery decreases.

Reduced oxygen delivery
→ decreased oxidative phosphorylation
→ decreased ATP production
→ increased anaerobic glycolysis
→ lactate production increases

3.3 Iron Metabolism

Iron is necessary for hemoglobin synthesis.

Normal mechanism:

Dietary iron absorbed in small intestine
→ iron transported in blood by transferrin

→ iron stored as ferritin
→ iron incorporated into hemoglobin during erythropoiesis

Without adequate iron:

Hemoglobin synthesis decreases
→ RBC oxygen-carrying capacity decreases
→ microcytic hypochromic anemia develops

3.4 Iron Deficiency Anemia ^^

Iron deficiency anemia is the most common anemia worldwide. ^^

Mechanism:

Chronic blood loss
or
Poor iron intake
or
Malabsorption
→ iron stores decrease
→ hemoglobin synthesis impaired
→ smaller RBCs produced
→ oxygen delivery decreases

Because hemoglobin decreases:

RBCs become:

- microcytic ^^
- hypochromic ^^

Microcytic means small RBCs.

Hypochromic means decreased hemoglobin concentration.

Common causes include:

- chronic GI bleeding ^^
- menstruation
- poor dietary intake
- peptic ulcer disease
- colon cancer ^^

Reduced oxygen delivery produces:

- fatigue ^^
- weakness
- pallor ^^
- dyspnea
- tachycardia

3.5 Ferritin and Transferrin

Ferritin stores iron.

Iron deficiency
→ ferritin decreases ^^

Transferrin transports iron through blood.

Iron deficiency
→ transferrin production increases
→ total iron-binding capacity increases ^^

The body increases transferrin in an attempt to capture more available iron.

3.6 Megaloblastic Anemia ^^

Megaloblastic anemia develops when DNA synthesis becomes impaired. ^^

Major causes include:

- vitamin B12 deficiency ^^
- folate deficiency ^^

DNA synthesis impairment
→ nuclear maturation slows
→ RBC precursor division impaired
→ abnormally large RBCs develop

RBCs become:

- macrocytic ^^

- megaloblastic ^^

Because rapidly dividing marrow cells cannot divide normally:

Ineffective erythropoiesis develops
→ anemia occurs

3.6A Vitamin B12 Deficiency ^^

Vitamin B12 is necessary for DNA synthesis and neurologic function. ^^

Normal mechanism:

Dietary B12 binds intrinsic factor in stomach
→ B12 absorbed in terminal ileum
→ normal DNA synthesis maintained

Pernicious anemia develops when intrinsic factor decreases.

Autoimmune destruction of parietal cells
→ intrinsic factor decreases
→ B12 absorption impaired
→ megaloblastic anemia develops ^^

Neurologic injury may also occur.

B12 deficiency
→ abnormal myelin maintenance
→ neurologic dysfunction develops ^^

Possible findings include:

- paresthesias ^^
- gait abnormalities
- neuropathy
- cognitive changes

3.6B Folate Deficiency ^^

Folate is also necessary for DNA synthesis. ^^

Folate deficiency
→ impaired DNA synthesis
→ macrocytic anemia develops ^^

Unlike B12 deficiency:

Folate deficiency usually does not cause severe neurologic deficits.

Causes include:

- malnutrition
- alcoholism ^^
- malabsorption
- pregnancy

Alcoholism contributes because poor nutrition and impaired absorption decrease folate availability.

3.7 Hemolytic Anemia ^^

Hemolytic anemia develops when RBC destruction exceeds RBC production. ^^

Mechanism:

RBC destruction increases
→ hemoglobin released
→ bilirubin production increases
→ jaundice may develop
→ oxygen-carrying capacity decreases

Bone marrow attempts compensation.

Hemolysis
→ erythropoietin increases
→ reticulocyte production increases

3.7A Bilirubin Metabolism and Jaundice ^^

Hemoglobin breakdown releases heme.

Heme metabolism
→ unconjugated bilirubin produced
→ bilirubin transported to liver
→ conjugation occurs
→ bilirubin excreted into bile

Excessive hemolysis
→ bilirubin production increases
→ jaundice develops ^^

Bilirubin accumulation causes yellow discoloration of:

- skin
- sclera
- mucous membranes

3.8 Sickle Cell Disease ^^

Sickle cell disease is caused by abnormal hemoglobin S formation. ^^

Mechanism:

Beta-globin mutation develops
→ hemoglobin S produced
→ low oxygen conditions trigger hemoglobin polymerization
→ RBC deformation develops
→ sickled RBCs form

Sickled RBCs become rigid.

Rigid RBCs
→ microvascular obstruction develops
→ tissue ischemia develops
→ severe pain occurs ^^

Microvascular obstruction
→ infarction risk increases
→ organ injury develops

Repeated sickling also damages RBC membranes.

Membrane damage
→ hemolysis develops
→ chronic anemia develops ^^

Common findings include:

- pain crises ^^
- anemia ^^
- jaundice
- splenic dysfunction
- infection risk increases

Splenic injury develops because repeated infarction damages splenic tissue.

3.9 Thalassemia ^^

Thalassemia involves decreased globin-chain production. ^^

Mechanism:

Reduced globin synthesis
→ defective hemoglobin formation
→ ineffective erythropoiesis develops
→ microcytic anemia develops ^^

Because hemoglobin synthesis decreases:

RBCs become small and fragile.

Severe forms may produce:

- marrow expansion
- skeletal deformities
- splenomegaly

3.10 Aplastic Anemia ^^

Aplastic anemia is bone marrow failure. ^^

Mechanism:

Stem-cell destruction develops
→ marrow cellularity decreases
→ RBC production decreases
→ WBC production decreases
→ platelet production decreases

Pancytopenia develops.

Pancytopenia means reduction of:

- RBCs
- WBCs
- platelets

Consequences include:

- fatigue from anemia ^^
- infection risk from leukopenia ^^
- bleeding risk from thrombocytopenia ^^

Causes may include:

- radiation ^^
- chemotherapy ^^
- toxins
- autoimmune marrow injury

3.11 Polycythemia ^^

Polycythemia means increased RBC concentration. ^^

Mechanism:

Excess erythrocyte production develops
→ blood viscosity increases
→ circulation slows
→ thrombosis risk increases ^^

Primary polycythemia may result from marrow mutation.

Secondary polycythemia may result from chronic hypoxia.

Chronic hypoxia
→ erythropoietin increases
→ RBC production increases

Common causes include:

- COPD ^^
- high altitude ^^
- chronic hypoxemia

Increased blood viscosity may produce:

- headache
- hypertension
- thrombosis
- stroke risk

3.12 Blood Loss and Hemorrhagic Anemia ^^

Acute hemorrhage
→ circulating blood volume decreases
→ oxygen delivery decreases
→ hypovolemic compensation develops

Compensatory responses include:

- tachycardia
- vasoconstriction
- increased sympathetic activation

Severe hemorrhage
→ tissue perfusion decreases
→ shock may develop ^^

Chronic blood loss
→ iron depletion develops
→ iron deficiency anemia occurs ^^

TLDR: Part 4 covers platelet disorders, coagulation pathways, thrombosis, emboli, DIC, anticoagulation, and bleeding-clotting pathophysiology.

ANSWER:

4. Disorders of Platelets, Coagulation, and Thrombosis

Normal hemostasis prevents excessive bleeding while maintaining blood flow through vessels.

Hemostasis occurs through two major processes:

- primary hemostasis
- secondary hemostasis

Primary hemostasis forms the platelet plug.

Secondary hemostasis stabilizes the platelet plug using fibrin.

Disorders develop when:

- platelets become abnormal
- coagulation factors become deficient
- inappropriate clotting develops
- excessive bleeding develops

4.1 Primary Hemostasis

Primary hemostasis forms the initial platelet plug.

Mechanism:

Endothelial injury
→ subendothelial collagen exposed
→ von Willebrand factor binds collagen
→ platelet adhesion occurs
→ platelet activation occurs
→ platelet aggregation occurs
→ temporary platelet plug forms

Activated platelets release:

- ADP
- thromboxane A2
- calcium

ADP recruits additional platelets.

Thromboxane A2 promotes platelet activation and vasoconstriction.

GpIIb/IIIa receptors bind fibrinogen between platelets.

Fibrinogen bridging
→ platelet aggregation develops
→ platelet plug strengthened

4.2 Secondary Hemostasis

Secondary hemostasis stabilizes the platelet plug using fibrin.

The coagulation cascade contains:

- intrinsic pathway
- extrinsic pathway
- common pathway

Both pathways activate Factor X.

Factor X activation
→ thrombin generated
→ fibrinogen converted into fibrin
→ stable clot forms

4.3 Intrinsic Pathway ^^

The intrinsic pathway begins when blood contacts exposed collagen. ^^

Mechanism:

Collagen exposure
→ Factor XII activated
→ Factor XI activated

→ Factor IX activated
→ Factor VIII activated
→ Factor X activated

PTT measures intrinsic pathway function. ^^

Intrinsic pathway disorders commonly prolong PTT. ^^

4.4 Extrinsic Pathway ^^

The extrinsic pathway begins with tissue factor exposure. ^^

Mechanism:

Tissue injury
→ tissue factor released
→ Factor VII activated
→ Factor X activated

PT measures extrinsic pathway function. ^^

Extrinsic pathway disorders commonly prolong PT. ^^

4.5 Common Pathway ^^

Both intrinsic and extrinsic pathways converge at Factor X. ^^

Mechanism:

Factor X activation
→ prothrombin converted into thrombin
→ fibrinogen converted into fibrin
→ fibrin cross-linking occurs
→ stable clot develops

Thrombin is central because it converts soluble fibrinogen into insoluble fibrin strands.

4.6 Platelet Disorders

Platelet disorders impair primary hemostasis.

Platelet dysfunction
or
Low platelet count
→ impaired platelet-plug formation develops
→ mucosal and superficial bleeding increase

Platelet disorders commonly produce:

- petechiae ^^
- purpura ^^
- mucosal bleeding ^^
- epistaxis
- gingival bleeding

Petechiae are small pinpoint hemorrhages caused by capillary bleeding.

Purpura are larger hemorrhagic skin lesions.

4.7 Thrombocytopenia ^^

Thrombocytopenia means decreased platelet count. ^^

Mechanism:

Decreased platelet production
or
Increased platelet destruction
or
Platelet consumption
→ platelet numbers decrease
→ bleeding risk increases

Common causes include:

- bone marrow failure ^^
- chemotherapy ^^
- immune destruction
- DIC ^^

- hypersplenism

Low platelet count
→ impaired primary hemostasis
→ petechiae and mucosal bleeding develop ^^

4.8 Immune Thrombocytopenic Purpura (ITP) ^^

ITP is autoimmune platelet destruction. ^^

Mechanism:

Autoantibodies bind platelets
→ splenic macrophages destroy platelets
→ thrombocytopenia develops
→ bleeding risk increases

Common findings include:

- petechiae ^^
- purpura ^^
- mucosal bleeding
- low platelet count ^^

Coagulation pathways are usually normal because the problem is platelet destruction rather than clotting-factor deficiency.

4.9 Hemophilia ^^

Hemophilia is coagulation-factor deficiency causing impaired secondary hemostasis. ^^

Hemophilia A involves Factor VIII deficiency. ^^

Mechanism:

Factor VIII deficiency
→ intrinsic pathway impaired
→ thrombin generation decreases

→ fibrin formation decreases
→ stable clot formation impaired
→ prolonged bleeding develops ^^

PTT becomes prolonged because the intrinsic pathway is impaired. ^^

Deep tissue bleeding commonly occurs.

Findings may include:

- hemarthrosis ^^
- muscle hematomas
- prolonged bleeding after procedures

Hemarthrosis means bleeding into joints.

4.10 von Willebrand Disease ^^

von Willebrand disease involves defective von Willebrand factor. ^^

Normal vWF function:

- binds collagen
- mediates platelet adhesion
- stabilizes Factor VIII

Defective vWF
→ platelet adhesion impaired
→ primary hemostasis impaired
→ bleeding tendency develops

Because vWF stabilizes Factor VIII:

PTT may also become prolonged. ^^

Common findings include:

- mucosal bleeding ^^
- epistaxis
- menorrhagia
- prolonged bleeding time ^^

4.11 Hypercoagulability and Thrombosis ^^

Thrombosis means pathologic clot formation within vessels. ^^

Thrombus formation commonly follows Virchow triad:

- endothelial injury ^^
- abnormal blood flow/stasis ^^
- hypercoagulability ^^

These three factors strongly increase thrombosis risk.

4.11A Endothelial Injury ^^

Endothelial injury exposes subendothelial collagen and tissue factor.

Endothelial injury
→ platelet activation develops
→ coagulation activated
→ thrombus formation risk increases

Common causes include:

- hypertension ^^
- smoking ^^
- atherosclerosis ^^
- trauma

4.11B Stasis ^^

Normal blood flow helps prevent clot formation.

Stasis develops when blood flow slows.

Reduced blood flow
→ clotting factors accumulate
→ platelet-endothelium interaction increases
→ thrombosis risk increases

Common causes include:

- prolonged immobility ^^
- surgery ^^
- heart failure
- atrial fibrillation ^^

4.11C Hypercoagulability ^^

Hypercoagulability means increased tendency toward clot formation. ^^

Causes may include:

- malignancy ^^
- pregnancy
- oral contraceptives ^^
- inherited thrombophilia
- inflammation

Hypercoagulability
→ coagulation activation increases
→ thrombosis risk increases

4.12 Deep Vein Thrombosis (DVT) ^^

DVT is thrombus formation within deep veins, usually in the legs. ^^

Mechanism:

Stasis
or
Hypercoagulability
or
Endothelial injury
→ venous thrombus forms
→ venous obstruction develops
→ swelling and pain occur

Common findings include:

- unilateral leg swelling ^^
- warmth
- pain
- erythema

Major danger:

Venous thrombus dislodges
→ embolus forms
→ pulmonary embolism may occur ^^

4.13 Pulmonary Embolism (PE) ^^

Pulmonary embolism occurs when embolic material obstructs pulmonary arteries. ^^

Most pulmonary emboli originate from DVTs.

Mechanism:

DVT develops
→ thrombus dislodges
→ embolus travels through venous circulation
→ embolus reaches pulmonary arteries
→ pulmonary blood flow obstructed
→ ventilation-perfusion mismatch develops
→ hypoxemia develops

Large PE
→ right ventricular strain develops
→ hemodynamic collapse may occur

Common findings include:

- sudden dyspnea ^^
- pleuritic chest pain ^^
- tachycardia
- hypoxemia
- hemoptysis

4.14 Disseminated Intravascular Coagulation (DIC) ^^

DIC involves widespread pathologic coagulation activation. ^^

Mechanism:

Severe systemic trigger develops
→ massive coagulation activation occurs
→ widespread microthrombi form
→ clotting factors consumed
→ platelets consumed
→ secondary bleeding develops

Common triggers include:

- sepsis ^^
- severe trauma ^^
- obstetric complications
- malignancy

DIC produces both:

- thrombosis
  and
- bleeding

Microvascular thrombosis
→ organ ischemia develops

Consumption of clotting factors
→ hemorrhage develops

Common findings include:

- bleeding from multiple sites ^^
- petechiae ^^
- organ dysfunction
- thrombocytopenia ^^
- prolonged PT/PTT ^^

4.15 Anticoagulants and Antiplatelet Drugs ^^

Anticoagulants decrease fibrin formation.

Antiplatelet drugs impair platelet activation.

Heparin enhances antithrombin activity.

Warfarin inhibits vitamin K-dependent clotting-factor synthesis.

Aspirin inhibits thromboxane A2 production.

Reduced thromboxane A2
→ platelet activation decreases
→ platelet aggregation decreases ^^

These medications reduce thrombosis risk but increase bleeding risk.

---

5. Shock

Shock is a state of inadequate tissue perfusion causing cellular hypoxia and organ dysfunction. ^^

Normal tissue survival requires:

- adequate blood volume
- adequate cardiac output
- adequate vascular tone
- adequate oxygen delivery

Shock develops when circulation can no longer maintain sufficient tissue perfusion.

Inadequate perfusion
→ decreased oxygen delivery
→ decreased oxidative phosphorylation
→ decreased ATP production
→ anaerobic metabolism increases
→ lactate production increases
→ cellular dysfunction develops
→ organ failure may occur

Regardless of cause, prolonged shock eventually causes:

- cellular injury
- membrane dysfunction
- inflammatory activation
- organ failure
- death

5.1 Compensatory Responses in Shock

The body initially attempts compensation.

Decreased tissue perfusion
→ sympathetic nervous system activated
→ catecholamine release increases
→ heart rate increases
→ contractility increases
→ peripheral vasoconstriction develops

Reduced renal perfusion
→ renin release increases
→ RAAS activated
→ sodium and water retention increase
→ blood pressure support attempted

Reduced perfusion also stimulates ADH release.

ADH
→ water retention increases
→ circulating volume partially supported

Early compensation attempts to preserve perfusion to vital organs:

- brain
- heart

Peripheral vasoconstriction diverts blood away from skin and extremities.

5.2 Stages of Shock

5.2A Compensated Shock

Compensation initially maintains blood pressure.

Possible findings include:

- tachycardia
- delayed capillary refill
- anxiety
- mild oliguria

Cellular hypoxia has begun, but compensation still partially preserves organ perfusion.

5.2B Progressive Shock

Compensation begins failing.

Persistent hypoperfusion
→ worsening anaerobic metabolism develops
→ metabolic acidosis worsens
→ myocardial function decreases
→ organ dysfunction progresses

Possible findings include:

- hypotension ^^
- confusion
- oliguria
- elevated lactate

5.2C Irreversible Shock

Severe prolonged hypoperfusion
→ widespread cellular injury develops
→ membrane failure occurs
→ inflammatory cascades amplify
→ multiorgan failure develops

Possible findings include:

- severe hypotension
- coma
- renal failure
- cardiovascular collapse

At this stage, recovery becomes unlikely even if circulation is restored.

5.3 Hypovolemic Shock ^^

Hypovolemic shock develops from severe loss of circulating volume. ^^

Major causes include:

- hemorrhage ^^
- severe dehydration
- burns

Mechanism:

Blood or fluid loss
→ venous return decreases
→ preload decreases
→ stroke volume decreases
→ cardiac output decreases
→ tissue perfusion decreases
→ cellular hypoxia develops

Hemorrhage additionally reduces oxygen-carrying capacity.

Reduced RBC mass
→ oxygen delivery worsens further
→ tissue hypoxia increases

Common findings include:

- tachycardia
- hypotension ^^
- weak pulses
- oliguria
- altered mental status

5.4 Cardiogenic Shock ^^

Cardiogenic shock develops when the heart cannot pump effectively. ^^

Major causes include:

- myocardial infarction ^^
- severe arrhythmia
- severe heart failure ^^

Mechanism:

Myocardial pump failure develops
→ cardiac output decreases
→ tissue perfusion decreases
→ compensatory vasoconstriction increases afterload
→ myocardial workload worsens
→ shock progresses

In myocardial infarction:

Coronary artery occlusion
→ myocardial ischemia develops
→ ATP depletion occurs
→ myocardial contractility decreases
→ cardiac output falls
→ cardiogenic shock may develop ^^

Left ventricular dysfunction may also produce pulmonary edema.

Left ventricular failure
→ pulmonary venous pressure increases
→ fluid enters alveoli
→ pulmonary edema develops ^^

Common findings include:

- hypotension ^^
- dyspnea ^^
- pulmonary edema ^^
- jugular venous distention ^^

5.5 Distributive Shock ^^

Distributive shock develops from severe pathologic vasodilation. ^^

Mechanism:

Systemic vasodilation develops
→ systemic vascular resistance decreases
→ blood pressure falls
→ tissue perfusion becomes inadequate

Major distributive shock subtypes include:

- septic shock ^^
- anaphylactic shock ^^
- neurogenic shock ^^

Unlike hypovolemic shock:

Circulating volume may initially be normal.

The major problem is loss of vascular tone.

5.5A Septic Shock ^^

Septic shock develops from overwhelming systemic infection and inflammatory activation. ^^

Mechanism:

Severe infection develops
→ macrophages activated
→ TNF-alpha and IL-1 release increase
→ nitric oxide production increases
→ widespread vasodilation develops
→ vascular permeability increases
→ hypotension develops ^^

Inflammatory injury also activates coagulation pathways.

Microvascular thrombosis develops
→ tissue perfusion worsens
→ organ dysfunction develops

Increased vascular permeability
→ intravascular fluid leaves circulation
→ relative hypovolemia develops
→ hypotension worsens

Common findings include:

- fever ^^
- hypotension ^^
- elevated lactate
- multiorgan dysfunction

Septic shock may progress to DIC. ^^

5.5B Anaphylactic Shock ^^

Anaphylactic shock is severe systemic Type I hypersensitivity causing distributive shock. ^^

Mechanism:

Systemic allergen exposure
→ widespread mast-cell degranulation occurs
→ histamine release massively increases
→ vasodilation develops
→ vascular permeability increases
→ hypotension develops
→ bronchoconstriction develops
→ airway edema develops ^^

Fluid leaves vessels because permeability increases.

Intravascular volume decreases
→ perfusion worsens
→ shock develops

Common findings include:

- hypotension ^^
- urticaria ^^
- wheezing ^^
- airway edema ^^

Epinephrine treatment:

Epinephrine
→ vasoconstriction increases blood pressure
→ bronchodilation improves airflow ^^

5.5C Neurogenic Shock ^^

Neurogenic shock develops from loss of sympathetic vascular tone. ^^

Major causes include:

- spinal cord injury ^^

Mechanism:

Sympathetic signaling interrupted
→ vascular tone decreases
→ vasodilation develops
→ blood pressure falls

Unlike most shock states:

Bradycardia may occur. ^^

Reason:

Sympathetic stimulation decreases
→ heart rate fails to compensate appropriately

Common findings include:

- hypotension ^^
- bradycardia ^^
- warm dry skin ^^

5.6 Obstructive Shock ^^

Obstructive shock develops when mechanical obstruction impairs circulation. ^^

Major causes include:

- pulmonary embolism ^^
- cardiac tamponade ^^
- tension pneumothorax ^^

5.6A Pulmonary Embolism and Obstructive Shock ^^

Large pulmonary embolus
→ pulmonary arterial obstruction develops
→ right ventricular afterload increases
→ right ventricular output decreases
→ left ventricular filling decreases
→ cardiac output decreases
→ shock develops ^^

Common findings include:

- sudden dyspnea ^^
- chest pain ^^
- hypoxemia
- hypotension ^^

5.6B Cardiac Tamponade ^^

Fluid accumulates within pericardial sac
→ cardiac compression develops
→ ventricular filling impaired
→ stroke volume decreases
→ cardiac output decreases
→ obstructive shock develops ^^

Common findings include:

- hypotension ^^
- jugular venous distention ^^
- muffled heart sounds ^^

This combination is Beck triad. ^^

5.6C Tension Pneumothorax ^^

Air enters pleural space and cannot escape.

Progressive intrathoracic pressure increase
→ lung collapse develops
→ mediastinum shifts
→ venous return decreases
→ cardiac output decreases
→ obstructive shock develops ^^

Common findings include:

- hypotension ^^
- unilateral absent breath sounds ^^
- tracheal deviation ^^

5.7 Multiorgan Failure in Shock

Persistent hypoperfusion
→ cellular injury worsens
→ inflammatory cascades amplify
→ endothelial injury develops

→ microvascular thrombosis develops
→ organ failure progresses

Organs commonly affected include:

- kidneys
- lungs
- liver
- brain
- heart

Renal hypoperfusion
→ acute kidney injury develops ^^

Severe prolonged shock
→ multiorgan dysfunction syndrome develops
→ death may occur

TLDR: Yes, that makes sense.

ANSWER:

6. Chapter 15 — Arterial and Cardiovascular Disorders

Cardiovascular disease develops when blood vessels, blood pressure regulation, coronary perfusion, cardiac muscle function, or cardiac electrical conduction become abnormal.

Major mechanisms include:

- arterial wall injury
- plaque formation
- reduced blood flow
- myocardial ischemia
- myocardial infarction
- pump failure
- abnormal rhythm

6.1 Arteries

Arteries carry blood away from the heart under high pressure.

Normal arteries require:

- elastic vessel walls
- intact endothelium
- smooth muscle regulation
- open lumen
- normal blood pressure

Arterial disease develops when the vessel wall becomes damaged, stiffened, narrowed, inflamed, or weakened.

6.2 Arteriosclerosis ^^

Arteriosclerosis means hardening or stiffening of arteries. ^^

Mechanism:

Chronic arterial injury
→ vessel wall thickening develops
→ elasticity decreases
→ arterial compliance decreases
→ systolic pressure rises
→ cardiac workload increases

Stiffer arteries cannot expand normally during systole.

Reduced arterial elasticity
→ increased resistance to forward blood flow
→ increased afterload
→ increased strain on the left ventricle

6.3 Atherosclerosis ^^

Atherosclerosis is plaque formation within arterial walls. ^^

Atherosclerosis is a type of arteriosclerosis.

Mechanism:

Endothelial injury
→ LDL enters arterial wall
→ LDL becomes oxidized
→ macrophages ingest oxidized LDL
→ foam cells form
→ fatty streak develops
→ smooth muscle cells migrate
→ collagen and extracellular matrix deposited
→ fibrous plaque forms
→ arterial lumen narrows

As plaque enlarges:

Plaque growth
→ lumen narrowing
→ blood flow decreases
→ tissue oxygen delivery decreases
→ ischemia may develop ^^

6.4 LDL, HDL, and Plaque ^^

LDL contributes to plaque buildup in arteries. ^^

Mechanism:

High LDL cholesterol ^^
→ increased LDL entry into damaged endothelium
→ LDL oxidation
→ macrophage foam-cell formation
→ fatty streak formation
→ plaque progression
→ atherosclerosis risk increases ^^

HDL helps remove cholesterol from tissues and vessels.

Mechanism:

HDL participates in reverse cholesterol transport
→ cholesterol removal from peripheral tissues increases
→ plaque progression risk decreases

Poor diet ^^
especially diets high in saturated and trans fats
→ LDL levels may rise
→ plaque formation risk increases ^^

Physical inactivity ^^
→ worsened lipid metabolism
→ endothelial dysfunction risk increases
→ cardiovascular risk increases ^^

Age ^^
→ cumulative endothelial injury increases
→ arterial stiffness increases
→ atherosclerosis risk increases ^^

6.5 Hypertension ^^

Hypertension is chronically elevated blood pressure. ^^

Blood pressure depends mainly on:

- cardiac output
- systemic vascular resistance
- circulating blood volume

Mechanism:

Increased vascular resistance
or
increased circulating volume
or
increased cardiac output
→ arterial pressure rises
→ endothelial stress increases
→ arterial injury develops

Persistent hypertension
→ endothelial injury
→ LDL entry into arterial wall increases
→ atherosclerosis accelerates
→ vessel stiffness worsens
→ blood pressure rises further

This creates a cycle:

Hypertension
→ endothelial injury
→ arterial stiffening
→ increased resistance
→ worsening hypertension

Long-term hypertension also affects the heart.

Chronic pressure overload
→ left ventricle pumps against higher afterload
→ myocardial cells enlarge
→ left ventricular hypertrophy develops

Hypertrophy initially helps generate stronger contraction.

Over time:

Left ventricular hypertrophy
→ myocardial oxygen demand increases
→ ventricular stiffness increases

→ diastolic filling worsens
→ heart failure risk increases

6.6 Coronary Artery Disease ^^

Coronary artery disease develops when coronary arteries become narrowed by atherosclerotic plaque. ^^

Mechanism:

Coronary atherosclerosis
→ coronary lumen narrows
→ myocardial blood flow decreases
→ oxygen supply to myocardium decreases
→ myocardial ischemia develops ^^

The myocardium extracts a high amount of oxygen at baseline.

Therefore, when oxygen demand rises during exertion, the heart depends on increased coronary blood flow.

If coronary arteries are narrowed:

Exercise or stress
→ myocardial oxygen demand increases
→ coronary blood flow cannot increase enough
→ ischemia develops
→ chest pain occurs

6.7 Exercise Stress Test ^^

An exercise stress test evaluates coronary artery function during physical exertion. ^^

Mechanism:

Exercise
→ heart rate increases
→ contractility increases
→ myocardial oxygen demand increases

If coronary arteries are significantly narrowed:

Increased oxygen demand
→ inadequate coronary supply
→ ischemic ECG changes or chest pain may occur

Stress testing is useful because coronary disease may not appear at rest.

At rest:

Oxygen demand may be low enough
→ coronary narrowing may not cause symptoms

During exercise:

Oxygen demand rises
→ narrowed coronary arteries fail to meet demand
→ ischemia becomes detectable

6.8 Angina Pectoris ^^

Angina pectoris is chest pain caused by myocardial ischemia. ^^

Mechanism:

Coronary blood flow decreases
→ myocardial oxygen delivery decreases
→ ATP production decreases
→ anaerobic metabolism increases
→ lactic acid and metabolites accumulate
→ pain fibers stimulated
→ chest pain develops

Angina means ischemia without myocardial cell death.

If ischemia resolves before necrosis:

Chest pain improves
→ troponin remains normal
→ myocardial infarction has not occurred

6.9 Stable Angina ^^

Stable angina is predictable ischemic chest pain, usually triggered by exertion and relieved by rest or nitroglycerin. ^^

Mechanism:

Fixed coronary plaque narrows artery
→ resting blood flow may be adequate
→ exertion increases myocardial oxygen demand
→ narrowed vessel cannot increase flow enough
→ temporary ischemia develops
→ chest pain occurs

Rest relieves stable angina because:

Rest
→ heart rate decreases
→ contractility decreases
→ myocardial oxygen demand decreases
→ ischemia improves
→ chest pain resolves ^^

Nitroglycerin relieves stable angina because:

Nitroglycerin
→ venodilation
→ venous return decreases
→ preload decreases
→ ventricular wall stress decreases
→ myocardial oxygen demand decreases
→ ischemic pain improves ^^

Stable angina does not usually cause troponin elevation because myocardial necrosis does not occur.

6.10 ST Depression

ST depression is commonly associated with subendocardial ischemia.

Subendocardium is the inner layer of myocardium.

It is vulnerable because it is exposed to high intraventricular pressure and receives perfusion mainly during diastole.

Mechanism:

Partial coronary flow limitation
→ inner myocardium receives inadequate oxygen
→ subendocardial ischemia develops
→ repolarization abnormalities occur
→ ST depression may appear

ST depression suggests ischemia but not necessarily full-thickness myocardial injury.

It may be seen with:

- stable angina during exertion
- unstable angina
- NSTEMI

6.11 Unstable Angina ^^

Unstable angina is worsening myocardial ischemia, often occurring at rest or with minimal exertion. ^^

Mechanism:

Atherosclerotic plaque becomes unstable
→ plaque rupture or erosion occurs
→ platelet activation develops
→ partially occlusive thrombus forms
→ coronary blood flow suddenly decreases
→ ischemia becomes more severe and less predictable

Pain may occur:

- at rest ^^
- with minimal exertion
- with increasing frequency
- with increasing duration

Unstable angina is not reliably relieved by rest or nitroglycerin.

Reason:

The problem is no longer only increased oxygen demand over a fixed narrowing.

The problem is acute plaque disruption and thrombosis.

Unstable angina:

Ischemia present
→ no myocardial necrosis
→ troponin remains normal

ECG may show:

- ST depression
- T-wave inversion
- no diagnostic change

6.12 Acute Myocardial Infarction ^^

Acute myocardial infarction is myocardial necrosis caused by prolonged ischemia. ^^

Mechanism:

Atherosclerotic plaque ruptures
→ collagen and tissue factor exposed
→ platelets adhere and activate
→ coagulation cascade activates
→ thrombus enlarges
→ coronary artery occlusion worsens
→ oxygen delivery falls severely
→ ATP depletion progresses
→ ion pumps fail
→ calcium accumulates in cells
→ mitochondrial injury worsens
→ sarcolemma ruptures
→ myocardial necrosis occurs

Myocardial infarction differs from angina because infarction involves cell death.

Angina:
ischemia
→ no necrosis
→ troponin normal

MI:
ischemia persists
→ necrosis occurs
→ troponin rises ^^

6.13 MI Symptoms ^^

Common myocardial infarction symptoms include:

- severe crushing chest pain ^^
- diaphoresis ^^
- shortness of breath
- nausea
- radiation to jaw, neck, shoulder, or left arm
- Levine's sign ^^

Levine's sign refers to a patient clutching the chest with a fist to describe ischemic chest pain. ^^

Mechanism of diaphoresis:

Myocardial ischemia
→ sympathetic nervous system activation
→ catecholamine release
→ sweating develops

Elevated blood pressure may occur early from sympathetic activation, but hypotension may occur if pump failure develops.

6.14 Cardiac Troponin ^^

Cardiac troponin is released when myocardial cells are injured or necrotic. ^^

Mechanism:

Myocardial necrosis
→ cell membrane disruption
→ intracellular troponin leaks into blood
→ serum troponin rises ^^

Troponin is more specific for myocardial injury than hematocrit, HDL, or IgA.

Troponin elevation supports myocardial infarction when clinical findings match ischemia.

6.15 NSTEMI

NSTEMI usually results from incomplete coronary occlusion with subendocardial necrosis.

Mechanism:

Partially occlusive thrombus
→ severe reduction in coronary blood flow
→ subendocardial ischemia develops
→ myocardial necrosis occurs
→ troponin rises
→ ST elevation is absent

ECG pattern:

NSTEMI
→ ST depression or T-wave inversion may occur
→ no persistent ST elevation

Key distinction:

Unstable angina and NSTEMI may look similar clinically.

Unstable angina:
→ no necrosis
→ normal troponin

NSTEMI:
→ necrosis
→ elevated troponin

6.16 STEMI

STEMI usually results from complete coronary occlusion with transmural necrosis.

Mechanism:

Complete coronary thrombotic occlusion
→ full-thickness myocardial ischemia
→ transmural injury develops
→ myocardial necrosis occurs
→ troponin rises
→ ST elevation appears

Transmural means the full thickness of the ventricular wall is injured.

ST elevation reflects acute full-thickness myocardial injury.

STEMI is more likely to require urgent reperfusion.

6.17 Myocardial Ischemia vs Myocardial Infarction ^^

Myocardial ischemia means oxygen supply is inadequate for myocardial demand. ^^

Myocardial infarction means tissue death occurred due to prolonged ischemia. ^^

Sequence:

Coronary blood flow decreases
→ ischemia develops
→ ATP decreases
→ contractile dysfunction occurs
→ pain develops
→ prolonged ischemia causes necrosis
→ infarction develops

Ischemia can be reversible.

Infarction is irreversible tissue death.

6.18 Myocardial Apoptosis ^^

Myocardial apoptosis is programmed cell death. ^^

It is not the main mechanism of acute MI.

Acute MI is primarily necrosis due to ischemic injury.

Mechanism:

Coronary occlusion
→ severe hypoxia
→ ATP depletion

→ membrane failure
→ necrosis

Apoptosis may occur in some chronic remodeling contexts, but acute MI is defined by ischemic myocardial necrosis.

6.19 Tachycardia ^^

Tachycardia means abnormally increased heart rate. ^^

Tachycardia may worsen ischemia because:

Heart rate increases
→ myocardial oxygen demand increases
→ diastolic filling time decreases
→ coronary perfusion time decreases
→ ischemia worsens

Tachycardia may also occur as a compensatory response during pain, anxiety, blood loss, shock, or heart failure.

6.20 Coronary Artery Bypass Grafting — CABG ^^

CABG creates new routes for blood flow around coronary artery occlusions. ^^

Mechanism:

Severe coronary artery narrowing or blockage
→ myocardial perfusion impaired
→ graft vessel placed beyond blocked segment
→ blood bypasses occlusion
→ distal myocardial blood flow improves

Multiple bypasses can be done at one time. ^^

CABG can be used to treat severe coronary artery disease, angina, and myocardial infarction risk patterns. ^^

Common graft vessels may include:

- internal mammary artery
- radial artery
- saphenous vein

The purpose is not to dissolve plaque.

The purpose is to route blood around obstructed coronary segments.

6.21 Heart Failure ^^

Heart failure is a chronic condition in which the heart cannot pump effectively enough to meet body demands. ^^

Heart failure differs from myocardial infarction.

Heart failure:
→ chronic pump dysfunction
→ inadequate forward flow or congestion

Myocardial infarction:
→ acute ischemic event
→ myocardial necrosis from blocked coronary blood flow ^^

MI can cause heart failure if enough myocardium is damaged.

Mechanism:

Myocardial injury
→ contractility decreases
→ stroke volume decreases
→ cardiac output decreases
→ compensatory mechanisms activate
→ fluid retention and remodeling occur
→ heart failure develops

6.22 Orthopnea ^^

Orthopnea means shortness of breath when lying flat. ^^

Mechanism:

Lying flat
→ venous return to heart increases
→ failing left ventricle cannot handle increased preload
→ pulmonary venous pressure rises
→ fluid moves into lung interstitium/alveoli
→ breathing becomes difficult

Orthopnea is a hallmark symptom of heart failure. ^^

6.23 Left Ventricular Failure ^^

Left ventricular failure causes pulmonary congestion. ^^

Mechanism:

Left ventricle fails to eject blood effectively
→ blood backs up into left atrium
→ pulmonary venous pressure increases
→ pulmonary capillary hydrostatic pressure rises
→ fluid enters alveoli
→ pulmonary edema develops
→ gas exchange worsens

Clinical findings include:

- crackles in lungs ^^
- dyspnea
- orthopnea ^^
- pulmonary edema ^^

Crackles occur because fluid is present in alveoli and small airways.

6.24 Right Ventricular Failure ^^

Right ventricular failure causes systemic venous congestion. ^^

Mechanism:

Right ventricle fails to pump blood into pulmonary circulation
→ blood backs up into systemic veins
→ venous pressure increases
→ fluid accumulates in peripheral tissues and abdominal organs

Clinical findings include:

- jugular venous distention ^^
- peripheral edema
- hepatomegaly
- ascites ^^

JVD occurs because elevated right-sided pressure backs into the jugular veins.

Ascites occurs because systemic venous congestion increases hydrostatic pressure in abdominal venous circulation.

6.25 Cor Pulmonale ^^

Cor pulmonale is right-sided heart failure caused by pulmonary disease. ^^

Mechanism:

Chronic lung disease
→ pulmonary vasoconstriction and vascular remodeling
→ pulmonary hypertension develops
→ right ventricular afterload increases
→ right ventricle hypertrophies
→ right ventricular failure develops

Sequence:

Pulmonary disease
→ increased pulmonary vascular resistance
→ right ventricular strain
→ right ventricular failure
→ systemic venous congestion

6.25A Putting It All Together — Progressive Cardiovascular Failure

Chronic cardiovascular disease often progresses through connected stages rather than isolated disorders.

One major progression pathway is:

Hypertension ^^
and
atherosclerosis ^^
→ chronic increased afterload develops
→ left ventricle must generate higher pressure
→ left ventricular hypertrophy develops ^^

Left ventricular hypertrophy initially helps maintain cardiac output.

However:

Persistent pressure overload
→ myocardial stiffness increases
→ coronary oxygen demand increases
→ ventricular relaxation worsens
→ diastolic dysfunction develops

At the same time:

Coronary atherosclerosis ^^
→ coronary blood flow decreases
→ myocardial ischemia develops ^^
→ myocardial injury accumulates
→ myocardial contractility weakens

Over time:

Left ventricular dysfunction develops
→ left ventricular chamber may dilate
→ mitral valve annulus stretches
→ mitral regurgitation may develop

This is usually mitral regurgitation from ventricular dilation rather than primary mitral valve prolapse.

Mitral regurgitation
→ blood leaks backward into left atrium during systole
→ left atrial pressure increases

→ pulmonary venous pressure rises
→ pulmonary congestion develops

Pulmonary venous congestion
→ pulmonary capillary hydrostatic pressure increases
→ fluid enters lung tissue
→ pulmonary edema develops ^^

Persistent pulmonary vascular stress
→ pulmonary hypertension develops ^^

Pulmonary hypertension
→ right ventricle pumps against increased pulmonary resistance
→ right ventricular workload increases
→ right ventricular hypertrophy develops
→ right ventricular dilation may eventually occur

As the right ventricle dilates:

Tricuspid valve annulus stretches
→ tricuspid regurgitation may develop

Again, this is usually functional tricuspid regurgitation from ventricular dilation rather than primary tricuspid valve prolapse.

Progressive right ventricular dysfunction
→ right-sided heart failure develops ^^

Right-sided heart failure
→ systemic venous pressure increases
→ jugular venous distention develops ^^
→ hepatic congestion develops
→ ascites develops ^^
→ peripheral edema develops

Complete integrated sequence:

Hypertension ^^
and
atherosclerosis ^^
→ left ventricular hypertrophy ^^
→ left ventricular dysfunction
→ mitral regurgitation
→ pulmonary congestion
→ pulmonary hypertension
→ cor pulmonale ^^

→ right ventricular failure ^^
→ tricuspid regurgitation
→ JVD ^^
→ ascites ^^
→ systemic venous congestion

Important clarification:

Cor pulmonale classically refers specifically to right-sided heart failure caused by primary pulmonary disease such as COPD. ^^

In pure left-heart-failure progression, some instructors simply call this “secondary pulmonary hypertension with right-sided heart failure.”

However, clinically the physiologic sequence overlaps significantly.

6.26 Hypertrophic Cardiomyopathy ^^

Hypertrophic cardiomyopathy means the cardiac muscle is enlarged. ^^

Mechanism:

Abnormal myocardial hypertrophy develops
→ ventricular wall thickens
→ ventricular chamber may become smaller
→ diastolic filling worsens
→ outflow obstruction may occur
→ cardiac output may become limited during exertion

In athletes, this matters because intense exertion increases oxygen demand and can trigger arrhythmias in vulnerable myocardium.

Hypertrophic cardiomyopathy can increase risk of sudden cardiac death.

6.27 Dysrhythmias ^^

A dysrhythmia is an abnormal heart rhythm or irregular heartbeat. ^^

Normal rhythm requires organized electrical conduction:

SA node
→ atria depolarize
→ AV node delays signal
→ Bundle of His
→ bundle branches
→ Purkinje fibers
→ ventricles contract

Dysrhythmia develops when electrical impulse formation or conduction becomes abnormal.

Causes include:

- ischemia
- electrolyte imbalance
- myocardial scarring
- cardiomyopathy
- structural heart disease

6.28 Ventricular Fibrillation ^^

Ventricular fibrillation is rapid, uncoordinated quivering of the ventricles. ^^

Mechanism:

Chaotic ventricular electrical activity
→ coordinated ventricular contraction stops
→ stroke volume collapses
→ cardiac output falls to near zero
→ loss of consciousness and death may occur without defibrillation

VF is life-threatening because the ventricles are not pumping effectively.

6.29 Atrial Fibrillation ^^

Atrial fibrillation is chaotic atrial electrical activity. ^^

Mechanism:

Disorganized atrial impulses
→ ineffective atrial contraction
→ blood stasis in atria
→ thrombus risk increases
→ embolic stroke risk increases

Atrial fibrillation may also reduce ventricular filling efficiency.

6.30 Aneurysm ^^

An aneurysm is abnormal dilation of a blood vessel caused by vessel-wall weakness. ^^

Mechanism:

Chronic hypertension
or
atherosclerotic wall damage
or
connective tissue weakness
→ arterial wall weakens
→ vessel dilates
→ wall tension increases
→ rupture risk increases

Rupture can cause life-threatening hemorrhage.

Aortic aneurysm rupture
→ massive internal bleeding
→ hypovolemic shock
→ death may occur

6.31 Vasculitis ^^

Vasculitis means inflammation of blood vessels. ^^

Mechanism:

Immune-mediated vascular inflammation
→ endothelial injury
→ vessel-wall swelling

→ lumen narrowing
→ tissue perfusion decreases
→ ischemia develops

Vasculitis may cause:

- fever
- rash
- organ ischemia
- aneurysm formation
- thrombosis

6.32 Kawasaki Disease ^^

Kawasaki disease is a pediatric vasculitis affecting medium-sized arteries, especially coronary arteries. ^^

Mechanism:

Immune-mediated vascular inflammation
→ coronary artery wall inflammation
→ vessel-wall weakening
→ coronary aneurysm risk increases
→ myocardial ischemia or infarction risk may develop

Common findings may include:

- prolonged fever
- conjunctivitis
- mucous membrane changes
- rash
- swollen hands and feet
- cervical lymphadenopathy

Major concern:

Coronary artery aneurysm
→ thrombosis risk
→ myocardial ischemia risk

6.33 Cardiovascular Integration

The major sequence is:

Hyperlipidemia ^^
and hypertension ^^
→ endothelial injury
→ LDL entry into vessel wall ^^
→ plaque formation ^^
→ atherosclerosis ^^
→ coronary artery narrowing ^^
→ ischemia ^^
→ angina ^^

If plaque ruptures:

Plaque rupture
→ thrombus formation
→ acute coronary syndrome
→ unstable angina, NSTEMI, or STEMI

If ischemia persists:

Persistent ischemia
→ myocardial necrosis
→ myocardial infarction ^^
→ troponin elevation ^^

If myocardial damage is large:

MI
→ contractile dysfunction
→ heart failure ^^
or
cardiogenic shock

If electrical conduction becomes unstable:

Ischemia
→ dysrhythmia ^^
→ ventricular fibrillation ^^
→ sudden cardiac death

TLDR: Section 7 begins with respiratory physiology, ventilation/perfusion logic, hypoxemia, COPD, chronic bronchitis, emphysema, asthma, pneumonia, pulmonary edema, respiratory failure, and pulmonary vascular consequences.

ANSWER:

7. Respiratory Disorders

The respiratory system delivers oxygen to blood and removes carbon dioxide from the body.

Normal respiratory function requires:

- open airways
- effective ventilation
- intact alveoli
- normal pulmonary blood flow
- adequate gas exchange
- normal respiratory muscle function

Respiratory disease develops when airflow, gas exchange, pulmonary perfusion, or lung structure becomes abnormal.

7.1 Ventilation and Gas Exchange

Ventilation means movement of air into and out of the lungs.

Gas exchange occurs at the alveoli.

Normally:

Air reaches alveoli
→ oxygen diffuses across alveolar membrane
→ oxygen enters pulmonary capillaries
→ hemoglobin binds oxygen
→ oxygen delivered to tissues

At the same time:

Carbon dioxide diffuses from blood
→ enters alveoli
→ exhaled from body

Efficient gas exchange requires matching between:

- ventilation
  and
- perfusion

7.2 Ventilation/Perfusion Matching ^^

Ventilation/perfusion matching means alveolar airflow and pulmonary blood flow must be appropriately coordinated. ^^

Ventilation without perfusion:
→ oxygen enters alveoli
→ little blood available to receive oxygen
→ physiologic dead space develops

Perfusion without ventilation:
→ blood reaches alveoli
→ oxygen cannot adequately enter blood
→ shunting develops

Both situations impair oxygenation.

Ventilation/perfusion mismatch
→ arterial oxygen decreases
→ hypoxemia develops ^^

7.3 Hypoxemia ^^

Hypoxemia means decreased oxygen concentration in arterial blood. ^^

Causes include:

- V/Q mismatch ^^
- hypoventilation
- diffusion impairment
- shunting
- high altitude

Reduced arterial oxygen
→ tissue oxygen delivery decreases
→ cellular hypoxia develops
→ compensatory responses activate

Compensatory responses include:

- tachycardia
- increased respiratory rate
- sympathetic activation

Severe hypoxemia
→ confusion
→ cyanosis ^^
→ organ dysfunction
→ respiratory failure may develop

7.4 Cyanosis ^^

Cyanosis is bluish discoloration caused by increased deoxygenated hemoglobin. ^^

Mechanism:

Arterial oxygen decreases
→ deoxygenated hemoglobin increases
→ bluish discoloration develops in skin and mucosa

Common locations include:

- lips
- nail beds
- mucous membranes

Cyanosis suggests significant oxygenation impairment.

7.5 Chronic Obstructive Pulmonary Disease (COPD) ^^

COPD is chronic progressive airflow obstruction. ^^

Major COPD subtypes include:

- chronic bronchitis ^^
- emphysema ^^

Smoking ^^
is the major risk factor.

Chronic smoke exposure
→ airway inflammation develops
→ mucus production increases
→ alveolar injury develops
→ airflow obstruction progresses

COPD produces difficulty exhaling air.

Air trapping develops because obstructed airways collapse during expiration.

7.6 Chronic Bronchitis ^^

Chronic bronchitis involves chronic airway inflammation with excessive mucus production. ^^

Mechanism:

Chronic smoke or irritant exposure
→ bronchial epithelial injury develops
→ chronic inflammation occurs
→ goblet-cell hyperplasia develops
→ mucus production increases
→ airway narrowing develops
→ airflow obstruction occurs

Excess mucus
→ impaired airflow
→ impaired mucus clearance
→ infection risk increases ^^

Chronic productive cough develops because mucus accumulates in airways.

Chronic hypoventilation
→ carbon dioxide retention develops
→ hypoxemia develops ^^

Pulmonary vasoconstriction from chronic hypoxemia
→ pulmonary hypertension may develop
→ cor pulmonale risk increases ^^

7.7 Emphysema ^^

Emphysema involves destruction of alveolar walls and loss of elastic recoil. ^^

Mechanism:

Smoking-induced inflammation
→ protease activity increases
→ alveolar wall destruction develops
→ alveolar septa break down
→ elastic recoil decreases
→ air trapping develops

Normally:

Elastic recoil helps force air out during expiration.

Loss of recoil
→ expiration becomes difficult
→ lungs hyperinflate
→ gas exchange surface area decreases ^^

Destruction of alveolar walls
→ diffusion surface area decreases
→ oxygen transfer decreases
→ hypoxemia develops

Patients often compensate with rapid breathing to maintain oxygenation.

7.8 Asthma ^^

Asthma is reversible airway obstruction caused by airway hyperreactivity and inflammation. ^^

Mechanism:

Allergen or irritant exposure
→ mast-cell activation develops
→ inflammatory mediators released
→ bronchoconstriction develops
→ mucus production increases
→ airway edema develops
→ airflow obstruction occurs ^^

Bronchoconstriction narrows airways.

Narrowed airways
→ wheezing develops ^^
→ expiration becomes difficult
→ airflow decreases

Asthma commonly involves Type I hypersensitivity mechanisms. ^^

Important mediators include:

- histamine
- leukotrienes
- IgE ^^

Asthma differs from COPD because obstruction is usually reversible.

7.9 Status Asthmaticus ^^

Status asthmaticus is severe prolonged asthma unresponsive to usual treatment. ^^

Mechanism:

Severe bronchoconstriction develops
→ airflow obstruction worsens
→ ventilation decreases
→ carbon dioxide retention develops
→ respiratory fatigue occurs
→ respiratory failure risk increases

Severe obstruction
→ oxygenation worsens
→ hypoxemia develops
→ life-threatening respiratory compromise may occur

7.10 Pneumonia ^^

Pneumonia is infection and inflammation of lung parenchyma. ^^

Mechanism:

Pathogen enters lower respiratory tract
→ alveolar macrophages activated
→ cytokine release develops
→ neutrophils recruited
→ inflammatory exudate accumulates in alveoli
→ gas exchange impaired

Fluid-filled alveoli
→ oxygen diffusion decreases
→ hypoxemia develops ^^

Common findings include:

- fever ^^
- productive cough ^^
- crackles ^^
- dyspnea
- leukocytosis ^^

Consolidation develops because alveoli fill with inflammatory material rather than air.

7.11 Pulmonary Edema ^^

Pulmonary edema is fluid accumulation within alveoli and lung interstitium. ^^

Most commonly:

Left ventricular failure ^^
→ pulmonary venous pressure increases
→ hydrostatic pressure rises
→ fluid enters alveoli
→ pulmonary edema develops ^^

Fluid-filled alveoli impair oxygen exchange.

Pulmonary edema
→ oxygen diffusion decreases
→ dyspnea develops
→ crackles develop ^^

Severe pulmonary edema may produce:

- pink frothy sputum
- severe hypoxemia
- respiratory distress

7.12 Respiratory Acidosis ^^

Respiratory acidosis develops when carbon dioxide retention increases. ^^

Normally:

Carbon dioxide is exhaled through ventilation.

Hypoventilation
→ carbon dioxide retention develops
→ carbonic acid increases
→ blood pH decreases
→ respiratory acidosis develops ^^

Common causes include:

- COPD ^^
- severe asthma
- respiratory depression
- neuromuscular weakness

7.13 Pulmonary Hypertension ^^

Pulmonary hypertension means elevated pressure within pulmonary circulation. ^^

Mechanism:

Chronic hypoxemia
→ pulmonary vasoconstriction develops

→ pulmonary vascular resistance increases
→ pulmonary artery pressure rises

Major causes include:

- COPD ^^
- chronic lung disease
- recurrent pulmonary emboli
- left-sided heart failure ^^

Pulmonary hypertension increases right ventricular workload.

7.14 Cor Pulmonale ^^

Cor pulmonale is right-sided heart failure caused by pulmonary disease. ^^

Mechanism:

Chronic lung disease
→ chronic hypoxemia develops
→ pulmonary vasoconstriction occurs
→ pulmonary hypertension develops
→ right ventricular afterload increases
→ right ventricular hypertrophy develops
→ right ventricular failure occurs ^^

Right-sided failure causes:

- JVD ^^
- peripheral edema
- ascites ^^
- hepatic congestion

7.15 Pulmonary Embolism Review ^^

Pulmonary embolism obstructs pulmonary arterial blood flow. ^^

Mechanism:

DVT develops ^^
→ embolus dislodges
→ embolus travels to pulmonary arteries
→ pulmonary perfusion obstructed
→ ventilation/perfusion mismatch develops ^^
→ hypoxemia occurs
→ right ventricular strain develops

Large pulmonary emboli may produce:

- sudden dyspnea ^^
- pleuritic chest pain ^^
- tachycardia
- hypotension ^^

7.16 Respiratory Failure ^^

Respiratory failure occurs when the respiratory system cannot adequately oxygenate blood or remove carbon dioxide. ^^

Two major patterns exist:

Type I respiratory failure:
→ hypoxemic failure

Type II respiratory failure:
→ hypercapnic failure

Mechanism:

Severe lung dysfunction
→ gas exchange fails
→ oxygenation and/or ventilation become inadequate
→ organ dysfunction develops

Severe respiratory failure may require:

- mechanical ventilation
- oxygen support
- airway stabilization

7.17 Smoking Integration ^^

Smoking damages multiple respiratory defense systems simultaneously. ^^

Smoking
→ ciliary dysfunction develops
→ mucus clearance decreases
→ chronic inflammation develops
→ airway remodeling occurs
→ alveolar destruction develops
→ infection risk increases
→ COPD develops ^^

Smoking also contributes to:

- atherosclerosis ^^
- endothelial injury ^^
- lung cancer risk
- chronic hypoxemia
- pulmonary hypertension
- cor pulmonale ^^

7.18 Respiratory System Integration

Smoking ^^
→ chronic airway inflammation
→ chronic bronchitis ^^

Smoking ^^
→ alveolar destruction
→ emphysema ^^

COPD ^^
→ chronic hypoxemia
→ pulmonary vasoconstriction
→ pulmonary hypertension ^^
→ cor pulmonale ^^

Left-sided heart failure ^^
→ pulmonary edema ^^
→ impaired oxygen exchange
→ hypoxemia ^^

DVT ^^
→ pulmonary embolism ^^
→ V/Q mismatch ^^
→ hypoxemia ^^

Pneumonia ^^
→ alveolar exudate
→ impaired gas exchange
→ hypoxemia ^^

Asthma ^^
→ bronchoconstriction
→ airflow obstruction
→ wheezing ^^

TLDR: Section 8 below integrates the entire Module 2 packet into complete disease chains tying immunity, hematology, coagulation, cardiovascular disease, shock, and respiratory pathology together.

ANSWER:

8. Final Systems Integration

Pathophysiology is the study of how normal physiology becomes abnormal.

Nearly all diseases in Module 2 follow the same progression:

Initial trigger
→ molecular or cellular dysfunction
→ tissue injury
→ organ dysfunction
→ clinical findings
→ complications

Understanding disease means understanding the entire chain rather than memorizing isolated facts.

8.1 Oxygen Delivery Integration ^^

Normal tissue survival requires adequate oxygen delivery. ^^

Oxygen delivery depends on:

- adequate lung function
- adequate hemoglobin
- adequate circulation
- adequate cardiac output

Failure at any step may produce tissue hypoxia.

Examples:

Pneumonia ^^
→ alveoli fill with inflammatory exudate
→ oxygen diffusion decreases
→ hypoxemia develops ^^

COPD ^^
→ airflow obstruction develops

→ ventilation decreases
→ oxygenation decreases
→ chronic hypoxemia develops ^^

Iron-deficiency anemia ^^
→ hemoglobin decreases
→ oxygen-carrying capacity decreases
→ tissue oxygen delivery decreases ^^

Heart failure ^^
→ cardiac output decreases
→ tissue perfusion decreases
→ oxygen delivery decreases ^^

Shock ^^
→ perfusion decreases
→ oxygen delivery collapses
→ anaerobic metabolism develops ^^

8.2 Hypoxia and Anaerobic Metabolism

Cells normally produce ATP through aerobic metabolism.

Reduced oxygen delivery
→ oxidative phosphorylation decreases
→ ATP production decreases
→ anaerobic glycolysis increases
→ lactate production increases
→ metabolic acidosis develops

Persistent ATP depletion
→ ion-pump failure develops
→ sodium accumulates intracellularly
→ water enters cells
→ cellular swelling develops

Severe prolonged hypoxia
→ membrane failure occurs
→ necrosis develops

This sequence occurs in:

- myocardial infarction ^^

- shock ^^
- severe anemia
- respiratory failure ^^
- pulmonary embolism ^^

8.3 Inflammation Integration ^^

Inflammation is a protective response to injury or infection.

Tissue injury
or
infection
→ macrophages activated
→ cytokine release develops
→ vascular permeability increases
→ leukocyte recruitment occurs
→ tissue inflammation develops

Important cytokines include:

- TNF-alpha
- IL-1
- IL-6

Inflammation is necessary for defense but may also cause tissue injury.

Examples:

SLE ^^
→ immune-complex deposition
→ complement activation
→ neutrophil-mediated tissue injury develops

Rheumatoid arthritis ^^
→ chronic synovial inflammation
→ pannus formation
→ cartilage destruction develops

Pneumonia ^^
→ neutrophilic alveolar inflammation
→ gas exchange impaired

Septic shock ^^
→ overwhelming inflammatory cytokine release
→ systemic vasodilation develops
→ hypotension occurs ^^

8.4 Autoimmunity Integration ^^

The immune system must tolerate self-antigens.

Failure of tolerance
→ autoreactive lymphocytes survive
→ autoimmune disease develops ^^

Different autoimmune diseases damage different tissues.

Examples:

SLE ^^
→ immune complexes deposit in kidneys
→ glomerulonephritis develops
→ hypertension develops ^^

Rheumatoid arthritis ^^
→ synovial inflammation develops
→ pannus forms
→ joint destruction occurs

ITP ^^
→ autoantibodies target platelets
→ thrombocytopenia develops
→ petechiae and bleeding occur ^^

Pernicious anemia ^^
→ autoimmune destruction of parietal cells
→ intrinsic factor decreases
→ B12 absorption decreases
→ megaloblastic anemia develops ^^

8.5 Hematology Integration ^^

Bone marrow produces:

- RBCs
- WBCs
- platelets

Bone marrow dysfunction affects multiple systems simultaneously.

Aplastic anemia ^^
→ stem-cell failure develops
→ pancytopenia develops
→ anemia, infection risk, and bleeding risk occur ^^

Leukemia ^^
→ malignant leukocyte proliferation develops
→ marrow overcrowding occurs
→ normal hematopoiesis suppressed
→ anemia and thrombocytopenia develop

Multiple myeloma ^^
→ malignant plasma cells proliferate
→ osteoclast activation increases
→ lytic lesions and fractures develop ^^

8.6 Anemia Integration ^^

Anemia means decreased oxygen-carrying capacity. ^^

Different mechanisms produce different anemia types.

Iron deficiency anemia ^^
→ impaired hemoglobin synthesis
→ microcytic hypochromic RBCs develop ^^

B12 or folate deficiency ^^
→ impaired DNA synthesis
→ macrocytic megaloblastic anemia develops ^^

Hemolytic anemia ^^
→ RBC destruction increases
→ bilirubin production increases
→ jaundice develops ^^

Sickle cell disease ^^
→ hemoglobin S polymerizes
→ RBC sickling develops
→ vaso-occlusion occurs
→ ischemic pain develops ^^

Regardless of cause:

Anemia
→ tissue hypoxia develops
→ fatigue occurs ^^
→ tachycardia develops
→ compensatory responses activate

8.7 Coagulation Integration ^^

Hemostasis requires:

- platelets
- coagulation factors
- intact endothelium

Primary hemostasis disorders commonly produce superficial bleeding.

Examples:

ITP ^^
→ thrombocytopenia develops
→ petechiae and purpura occur ^^

von Willebrand disease ^^
→ platelet adhesion impaired
→ mucosal bleeding develops ^^

Secondary hemostasis disorders impair fibrin formation.

Hemophilia A ^^
→ Factor VIII deficiency
→ intrinsic pathway impaired
→ prolonged bleeding develops ^^

Hypercoagulability causes pathologic thrombosis.

Virchow triad:

- endothelial injury ^^
- stasis ^^
- hypercoagulability ^^

DVT ^^
→ thrombus forms in deep veins
→ pulmonary embolism risk develops ^^

8.8 Shock Integration ^^

Shock is inadequate tissue perfusion causing cellular hypoxia. ^^

Regardless of type:

Perfusion decreases
→ oxygen delivery decreases
→ ATP depletion develops
→ organ dysfunction occurs

Hypovolemic shock ^^
→ blood volume decreases
→ preload decreases
→ cardiac output falls

Cardiogenic shock ^^
→ pump failure develops
→ cardiac output decreases

Septic shock ^^
→ systemic vasodilation develops
→ vascular permeability increases
→ hypotension occurs ^^

Obstructive shock ^^
→ mechanical obstruction impairs circulation

Progressive shock
→ lactate rises
→ acidosis worsens
→ multiorgan failure develops ^^

8.9 Cardiovascular Integration ^^

Atherosclerosis ^^
→ coronary narrowing develops
→ myocardial ischemia develops ^^

Stable angina ^^
→ transient ischemia during exertion
→ chest pain relieved by rest or nitroglycerin ^^

Plaque rupture
→ thrombosis develops
→ acute coronary syndrome occurs ^^

Persistent ischemia
→ myocardial necrosis develops
→ myocardial infarction occurs ^^

Myocardial infarction ^^
→ troponin rises
→ contractility decreases
→ arrhythmia risk increases ^^

Large infarction
→ left ventricular dysfunction develops
→ pulmonary edema occurs ^^
→ heart failure develops ^^

Pulmonary hypertension
→ right ventricular strain develops
→ cor pulmonale occurs ^^

Right-sided failure
→ JVD develops ^^
→ ascites develops ^^

8.10 Respiratory Integration ^^

COPD ^^
→ chronic airflow obstruction develops
→ hypoxemia occurs ^^

Chronic bronchitis ^^
→ mucus hypersecretion develops
→ chronic productive cough occurs

Emphysema ^^
→ alveolar destruction develops
→ diffusion surface area decreases
→ oxygenation worsens

Asthma ^^
→ bronchoconstriction develops
→ wheezing occurs ^^

Pneumonia ^^
→ alveolar inflammatory exudate develops
→ oxygen diffusion impaired
→ hypoxemia develops ^^

Pulmonary embolism ^^
→ pulmonary perfusion obstructed
→ V/Q mismatch develops ^^

Chronic hypoxemia
→ pulmonary vasoconstriction develops
→ pulmonary hypertension occurs
→ cor pulmonale develops ^^

8.11 Final Integrated Master Sequence

Smoking ^^
→ endothelial injury ^^
→ atherosclerosis develops ^^

Smoking ^^
→ chronic airway inflammation develops
→ COPD develops ^^

Atherosclerosis ^^
→ coronary artery disease develops
→ myocardial ischemia develops ^^

Myocardial infarction ^^
→ left ventricular dysfunction develops
→ pulmonary edema develops ^^

Pulmonary edema
→ impaired oxygen exchange develops
→ hypoxemia worsens ^^

Chronic pulmonary hypoxemia
→ pulmonary hypertension develops
→ cor pulmonale develops ^^

Right-sided heart failure
→ JVD develops ^^
→ ascites develops ^^

At the same time:

Immobility ^^
or
hypercoagulability ^^
→ DVT develops ^^

DVT ^^
→ pulmonary embolism develops ^^

Pulmonary embolism ^^
→ obstructive shock may develop ^^

Persistent shock
→ multiorgan failure develops ^^

9. Module 3 Completion Addendum

The previous sections already covered the major foundations of:

- ischemia
- MI
- heart failure
- DVT
- PE
- COPD
- asthma
- pneumonia
- pulmonary edema
- pulmonary hypertension
- cor pulmonale

This addendum fills in the remaining Module 3 details that still need explicit coverage.

9.1 Acute Coronary Syndrome Review ^^

Acute coronary syndrome, or ACS, describes acute myocardial ischemia caused by sudden reduction in coronary blood flow. ^^

ACS includes:

- unstable angina ^^
- NSTEMI
- STEMI
- acute myocardial infarction ^^

General sequence:

Atherosclerotic plaque
→ plaque instability
→ plaque rupture or erosion
→ platelet activation
→ thrombus formation
→ coronary blood flow decreases
→ myocardial ischemia develops
→ angina or myocardial infarction occurs

The difference between unstable angina and MI is myocardial necrosis.

Unstable angina:

Ischemia
→ no myocardial necrosis
→ troponin remains normal

Myocardial infarction:

Persistent ischemia
→ myocardial necrosis
→ troponin rises ^^

9.2 PTCA / Angioplasty ^^

Percutaneous transluminal coronary angioplasty, or PTCA, is a procedure used to open narrowed coronary arteries. ^^

Mechanism:

Coronary artery narrowed by plaque
→ catheter inserted into coronary artery
→ balloon inflated at narrowed site
→ plaque compressed against vessel wall
→ lumen widens
→ coronary blood flow improves

Often, a stent is placed.

Stent placement
→ vessel kept open
→ recurrent narrowing risk decreases

PTCA is different from CABG.

PTCA:

Narrowed vessel
→ opened from inside using balloon/stent

CABG:

Blocked vessel
→ bypassed using graft vessel ^^

9.3 CABG vs PTCA ^^

CABG creates a new blood-flow route around blocked coronary arteries. ^^

PTCA opens the narrowed artery directly.

CABG:

Severe coronary obstruction
→ graft vessel attached before and after blockage
→ blood bypasses obstruction
→ downstream myocardium receives blood

PTCA:

Coronary narrowing
→ balloon/stent opens existing vessel
→ blood flows through original artery

Both improve myocardial perfusion.

Neither reverses the entire systemic disease process.

Underlying atherosclerosis still requires risk-factor control.

9.4 Infective Endocarditis ^^

Infective endocarditis is infection of the endocardial surface of the heart, usually involving heart valves. ^^

Mechanism:

Bacteria enter bloodstream
→ bacteria adhere to damaged valve or endocardium
→ platelet-fibrin deposits trap bacteria
→ vegetations form
→ valve structure becomes damaged

Vegetations are infected masses made of:

- bacteria
- fibrin
- platelets
- inflammatory material

Valve damage may cause regurgitation.

Valve infection
→ leaflet destruction
→ poor valve closure
→ backward blood flow develops

Fragments may embolize.

Vegetation fragment breaks off
→ embolus travels through bloodstream
→ infarction or infection may occur elsewhere

Common risk factors include:

- damaged valves
- prosthetic valves
- IV drug use
- prior endocarditis

9.5 Valvular Disease

Valves normally keep blood moving forward.

Valve disease causes abnormal forward flow or backward leakage.

Two major valve problems:

- stenosis
- regurgitation

Stenosis:

Valve opening narrowed
→ forward flow obstructed
→ chamber behind valve works harder

Regurgitation:

Valve fails to close
→ blood leaks backward
→ volume overload develops

9.6 Aortic Stenosis ^^

Aortic stenosis is narrowing of the aortic valve. ^^

Mechanism:

Aortic valve narrows
→ left ventricle must generate higher pressure to eject blood
→ left ventricular afterload increases
→ left ventricular hypertrophy develops
→ myocardial oxygen demand increases
→ ischemia risk increases

Severe aortic stenosis may cause:

- exertional chest pain
- syncope
- dyspnea
- heart failure

Sequence:

Aortic valve narrowing
→ outflow obstruction
→ pressure overload
→ LV hypertrophy
→ reduced cardiac output during exertion
→ symptoms develop

9.7 Mitral Regurgitation ^^

Mitral regurgitation means the mitral valve leaks backward from the left ventricle into the left atrium. ^^

Mechanism:

Mitral valve fails to close properly
→ blood flows backward into left atrium during systole
→ left atrial volume increases
→ pulmonary venous pressure rises
→ pulmonary congestion may develop

Causes include:

- ventricular dilation
- papillary muscle dysfunction after MI
- infective endocarditis
- rheumatic valve disease
- mitral valve prolapse

Papillary muscle dysfunction after MI:

MI damages papillary muscle
→ mitral valve support weakens
→ valve closure becomes incomplete
→ mitral regurgitation develops

9.8 Systolic vs Diastolic Heart Failure ^^

Heart failure may involve impaired contraction, impaired filling, or both. ^^

Systolic heart failure:

Ventricle cannot contract effectively
→ ejection fraction decreases
→ forward cardiac output decreases

Common causes:

- myocardial infarction ^^
- dilated cardiomyopathy
- chronic ischemic heart disease

Diastolic heart failure:

Ventricle cannot relax or fill properly
→ filling pressure increases
→ congestion develops despite preserved contraction

Common causes:

- hypertension ^^
- left ventricular hypertrophy
- hypertrophic cardiomyopathy

Comparison:

Systolic failure
→ weak pump

Diastolic failure
→ stiff pump

Both can cause pulmonary congestion and dyspnea.

9.9 Low-Output vs High-Output Heart Failure

Low-output heart failure occurs when the heart cannot pump enough blood forward.

Mechanism:

Pump dysfunction
→ cardiac output decreases
→ tissue perfusion decreases
→ fatigue, weakness, and organ hypoperfusion occur

Common causes:

- MI
- cardiomyopathy
- hypertension
- valve disease

High-output heart failure occurs when cardiac output is high but still insufficient for body demand.

Mechanism:

Systemic demand increases
or
systemic vascular resistance falls
→ heart pumps more than normal

→ chronic overload develops
→ heart eventually fails

Possible causes:

- severe anemia
- hyperthyroidism
- sepsis
- arteriovenous fistula

Key distinction:

Low-output failure
→ pump cannot generate enough output

High-output failure
→ body demand is abnormally high or vascular resistance is too low

---

9.10 Cardiomyopathy Types ^^

Cardiomyopathy means disease of the heart muscle. ^^

Major types include:

- dilated cardiomyopathy
- hypertrophic cardiomyopathy ^^
- restrictive cardiomyopathy

Dilated cardiomyopathy:

Ventricular dilation
→ weak contraction
→ systolic failure develops

Hypertrophic cardiomyopathy:

Myocardial thickening
→ impaired ventricular filling
→ possible outflow obstruction
→ exertional syncope or sudden death risk increases ^^

Restrictive cardiomyopathy:

Stiff ventricular walls
→ impaired filling
→ diastolic failure develops

9.11 Venous Insufficiency and Varicose Veins ^^

Venous insufficiency occurs when veins cannot return blood effectively back to the heart. ^^

Normal veins contain valves.

Valves prevent backward flow.

Mechanism:

Valve weakness or vein dilation
→ venous blood pools
→ venous pressure increases
→ veins enlarge and twist
→ varicose veins develop ^^

Varicose veins are dilated tortuous superficial veins. ^^

Common findings:

- visible enlarged veins
- leg heaviness
- aching
- swelling
- worse symptoms after prolonged standing

Venous insufficiency can also cause:

Venous pressure increases
→ fluid leaves capillaries
→ edema develops
→ skin changes may occur
→ venous ulcers may develop

9.12 DVT Risk Integration ^^

Deep vein thrombosis develops when clot formation occurs in deep veins. ^^

Major risk logic follows Virchow triad:

- stasis ^^
- endothelial injury ^^
- hypercoagulability ^^

Examples:

Hip replacement surgery ^^
→ endothelial injury + immobility
→ DVT risk increases

Fracture ^^
→ vascular injury + immobility
→ DVT risk increases

Pregnancy ^^
→ hypercoagulability + venous compression
→ DVT risk increases

Smoking ^^
→ endothelial injury + hypercoagulability
→ DVT risk increases

Obesity ^^
→ venous stasis + inflammation
→ DVT risk increases

DVT
→ thrombus may embolize
→ pulmonary embolism may develop ^^

TLDR: Your current Module 2–3 packet ends at section 9.12, so the addendum should continue as Section 9.13 onward.

ANSWER:

9.13 Pleural Effusion ^^

Pleural effusion is abnormal fluid accumulation within the pleural space. ^^

Mechanism:

Heart failure
or
infection
or
malignancy
or
hypoproteinemia
→ hydrostatic pressure increases
or
oncotic pressure decreases
or
vascular permeability increases
→ fluid enters pleural space
→ lung expansion decreases
→ ventilation impaired

Large pleural effusions compress lung tissue.

Lung compression
→ ventilation decreases
→ dyspnea develops
→ hypoxemia may occur ^^

Common findings include:

• decreased breath sounds ^^
• dullness to percussion
• dyspnea ^^
• pleuritic chest discomfort

Pleural effusion commonly occurs in:
• heart failure ^^
• pneumonia ^^

• malignancy
• pulmonary embolism ^^

9.14 Pneumothorax ^^

Pneumothorax occurs when air enters the pleural space. ^^

Mechanism:

Air enters pleural cavity
→ intrapleural pressure increases
→ lung recoil occurs
→ partial or complete lung collapse develops
→ ventilation decreases
→ hypoxemia may occur ^^

Common findings include:

• sudden dyspnea ^^
• unilateral absent breath sounds ^^
• pleuritic chest pain
• tachycardia

Tension pneumothorax:

Air enters pleural space
→ air cannot escape
→ intrathoracic pressure progressively increases
→ mediastinal shift develops
→ venous return decreases
→ cardiac output decreases
→ obstructive shock develops ^^

Common findings include:

• hypotension ^^
• tracheal deviation ^^
• unilateral absent breath sounds ^^
• severe respiratory distress

9.15 Community-Acquired Pneumonia Review ^^

Community-acquired pneumonia develops outside the hospital setting. ^^

Most common bacterial cause:

• Streptococcus pneumoniae ^^

Mechanism:

Bacterial pathogen enters lower respiratory tract
→ alveolar macrophages activated
→ cytokine release develops
→ neutrophils recruited
→ inflammatory exudate fills alveoli
→ consolidation develops
→ oxygen diffusion impaired
→ hypoxemia develops ^^

Common findings include:

• fever ^^
• productive cough ^^
• crackles ^^
• leukocytosis ^^
• dyspnea

Consolidation means alveoli are filled with inflammatory material instead of air.

9.16 Infective Endocarditis Prevention ^^

Poor dental hygiene
→ oral bacterial burden increases
→ transient bacteremia risk increases
→ damaged valves may become infected
→ infective endocarditis risk increases ^^

IV drug use:

Contaminated injection
→ bacteria enter bloodstream
→ bacteria adhere to valves

→ vegetations form
→ valve destruction develops ^^

Prevention strategies include:

• proper dental hygiene ^^
• avoiding IV drug use ^^
• prophylactic antibiotics in selected high-risk patients

9.17 Sickle Cell Disease Genetics ^^

Sickle cell disease is an autosomal recessive disorder caused by beta-globin mutation. ^^

Mechanism:

Both parents carry abnormal beta-globin allele
→ child inherits two abnormal alleles
→ hemoglobin S produced
→ sickling develops ^^

Low oxygen conditions
→ hemoglobin S polymerizes
→ RBC deformation develops
→ vaso-occlusion occurs
→ ischemic pain develops ^^

Common findings include:

• pain crises ^^
• chronic hemolytic anemia ^^
• jaundice
• splenic dysfunction
• infection risk increases

9.18 Iron Deficiency Anemia Treatment ^^

Iron deficiency anemia develops when iron availability becomes inadequate for hemoglobin synthesis. ^^

Mechanism:

Iron deficiency
→ hemoglobin synthesis impaired
→ microcytic hypochromic anemia develops ^^
→ oxygen delivery decreases

Common treatments include:

• oral ferrous sulfate ^^
• intravenous iron supplementation ^^
• vitamin C supplementation ^^

Vitamin C improves iron absorption within the gastrointestinal tract.

Vitamin B12 injections are used for vitamin B12 deficiency anemia rather than iron deficiency anemia. ^^

9.19 Final High-Yield Rapid Integration ^^

Atherosclerosis ^^
→ coronary artery narrowing develops
→ myocardial ischemia develops ^^

Plaque rupture
→ thrombus formation develops
→ ACS occurs ^^

Persistent ischemia
→ myocardial necrosis develops
→ troponin rises ^^
→ myocardial infarction occurs ^^

MI ^^
→ decreased contractility
→ pulmonary edema develops ^^
→ cardiogenic shock may occur ^^

DVT ^^
→ pulmonary embolism develops ^^
→ V/Q mismatch develops ^^
→ hypoxemia occurs ^^

COPD ^^
→ chronic hypoxemia develops

→ pulmonary hypertension occurs ^^
→ cor pulmonale develops ^^

Shock ^^
→ tissue perfusion decreases
→ anaerobic metabolism develops
→ lactate increases
→ organ dysfunction develops ^^